

Bioinformatics Course Project

The Core Principle

Throughout the semester, keep asking:

What biological question are we trying to answer?

The data, tools and computational methods should follow from that question.

Biological Question → Data → Method → Analysis → Biological Meaning

1. Project Overview

This course will be delivered through a project-based learning approach. Rather than completing a single project at the end of the semester, you will progressively develop a Bioinformatics research project as you learn and apply different concepts and techniques throughout the course.

You will work in groups of up to 6 students to investigate a biological problem using biological data and computational methods.

Your project should demonstrate how Bioinformatics can be used to answer a meaningful biological question through the process:

Biological Question → Biological Data → Computational Analysis → Results → Biological Interpretation

The project will develop progressively throughout the semester. The techniques and tools used will be introduced during the relevant course topics and applied to your project where appropriate.

2. Project Tracks

Each group will work within one of the following biological-data-based tracks:

Track 1: Genomics

Projects involving DNA, genomes, genetic variation, genes and related genomic data.

Possible areas include:

- Comparative analysis of genes or genomes
- Genetic variation
- Mutation analysis
- Conserved genomic regions
- Disease-associated genes
- Antibiotic resistance genes

Track 2: Transcriptomics

Projects involving RNA and gene-expression data.

Possible areas include:

- Gene expression analysis
- Differential gene expression
- Disease vs. healthy expression patterns
- Gene-expression-based classification
- Identification of potentially important genes

Track 3: Proteomics

Projects involving proteins, protein sequences, protein expression or protein characteristics.

Possible areas include:

- Protein sequence comparison
- Protein function prediction
- Conserved protein regions
- Protein families
- Disease-associated proteins
- Protein characteristics and classification

Track 4: Microbial Bioinformatics / Metagenomics

Projects involving microorganisms, microbial genomes or microbial communities.

Possible areas include:

- Bacterial genome comparison
- Pathogen identification
- Antibiotic resistance
- Microbial diversity
- Microbial community composition
- Comparative analysis of microbial strains

Track 5: Phylogenetics and Evolution

Projects investigating evolutionary relationships using biological sequence or other appropriate biological data.

Possible areas include:

- Evolutionary relationships between species
- Gene or protein evolution
- Phylogenetic tree construction
- Comparative sequence analysis
- Molecular evolution

Track 6: Applied / Clinical Bioinformatics

Projects applying Bioinformatics to biological or biomedical problems.

Possible areas include:

- Disease-associated genes or proteins
- Biomarker identification
- Drug targets
- Disease classification using biological data
- Biological prediction and classification
- Other computational analyses of biological datasets

Important: A project must primarily analyse **biological data**. Projects based primarily on fingerprints, iris images, facial recognition, generic image classification, or other biometric/computer-vision applications do not fall under this course.

3. Project Requirements

Every project must contain the following:

A. A Biological Problem

Identify a biological problem or area of interest that can be investigated using computational methods. The problem should have a clear biological context and should be sufficiently focused to be investigated within the semester.

B. A Research Question

Develop a specific question that your project will attempt to answer.

Your question should be:

- Clearly stated
- Biologically meaningful
- Specific enough to investigate
- Answerable using available biological data

C. Project Objectives

Define 2–4 specific objectives that describe what your project intends to accomplish.

Your objectives should directly contribute to answering your research question.

D. Defined Scope

Clearly state what your project will and will not investigate.

Consider:

- Organism(s)
- Gene(s), protein(s) or biological system(s)
- Dataset(s)
- Conditions or populations
- Time period, where relevant
- Analysis to be performed

A focused project is preferable to a project that attempts to investigate too much.

E. Biological Data

Identify the type of biological data required to answer your question.

Examples include:

- DNA sequences
- RNA sequences
- Gene-expression data
- Protein sequences
- Protein structures
- Genetic variants
- Microbial genomes

- Metagenomic data
- Biological interaction data

The specific datasets and databases to be used will be determined as the relevant course topics are covered.

F. Computational Analysis

Apply appropriate Bioinformatics techniques to your data.

Depending on your project, these may include:

- Data acquisition
- Data cleaning and preprocessing
- Exploratory data analysis
- Sequence alignment
- BLAST
- Feature extraction
- Feature selection
- Classification
- Machine learning
- Statistical analysis
- Phylogenetic analysis
- Other appropriate computational methods

You are not expected to use every technique taught in the course.

The methods selected should be appropriate for your biological question and dataset.

G. Biological Interpretation

Your project must go beyond producing computational results.

You must explain:

What do your results mean from a biological perspective?

Your conclusions should be supported by your analysis and should acknowledge relevant limitations.

4. Project Development

The project will be developed progressively throughout the semester.

Stage 1 - Define the Problem

At the beginning of the course, each group will submit:

- Group number
- Preferred project track
- Biological problem
- Research question
- Target beneficiaries/stakeholders
- Project scope
- Initial objectives
- Initial idea of the biological data required

At this stage, you **do not need to have a complete methodology**.

The purpose is to establish a direction for the project.

Stage 2 - Identify the Data

After learning about biological data and databases, groups will identify appropriate sources of data for their project.

You will:

- Identify appropriate biological datasets
- Identify relevant databases
- Examine the characteristics of the data
- Justify why the selected data can help answer your research question

Stage 3 - Prepare and Explore the Data

As data preparation and exploratory analysis techniques are introduced, you will apply them to your project.

This may include:

- Data cleaning
- Handling missing or inconsistent data
- Data transformation
- Descriptive statistics

- Exploratory visualisations
- Identifying patterns and relationships

Stage 4 - Perform Bioinformatics Analysis

You will apply the Bioinformatics techniques relevant to your project.

For example, a project involving DNA sequences may require sequence alignment and BLAST, while a gene-expression project may require statistical analysis, feature selection or machine learning.

The project methodology should be driven by the research question.

You should not use a technique simply because it was taught. You should be able to explain why a particular method is appropriate for your project.

Stage 5 - Interpret the Results

Analyse your findings and determine whether they provide an answer to your research question.

You should discuss:

- Major findings
- Biological significance
- Patterns observed
- Possible explanations
- Limitations of the analysis
- What could be investigated in future work

Stage 6 - Final Project Presentation

At the end of the semester, each group will present its completed project as a **mini-Bioinformatics research study**.

The presentation should demonstrate the complete journey of your project.

5. Project Checkpoints

The project will be assessed progressively through checkpoints rather than as one final submission.

Checkpoints	Focus
CP 1	Biological problem, research question, scope and objectives
CP 2	Biological data and data sources
CP 3	Data preparation and exploratory analysis
CP 4	Bioinformatics analysis and techniques
CP 5	Results and biological interpretation
Final	Complete project presentation/report

Further instructions, requirements and deadlines for each checkpoint will be provided as the relevant course topics are covered.

6. Group Responsibilities

Each group member is expected to contribute meaningfully to the project.

Groups should establish responsibilities for areas such as:

- Project coordination
- Data acquisition and preparation
- Computational analysis
- Biological interpretation
- Visualisations
- Documentation and presentation

Responsibilities may be rotated throughout the project.

Every group member must understand the overall project and be able to explain the group's methodology and findings.

7. Project Expectations

Your project should demonstrate:

- Understanding of the biological problem
- Appropriate selection and use of biological data
- Correct application of Bioinformatics concepts
- Appropriate computational analysis
- Clear presentation of results
- Ability to interpret computational results biologically
- Evidence-based conclusions
- Effective teamwork
- Appropriate use and acknowledgement of scientific sources

The goal is **not simply to produce code, graphs or computational outputs.**

The goal is to use computational methods to **answer a biological question.**